GENOME PODS

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Scalable Personalized Medicine?

€ Unfortunately, still quite expensive

But why? Sequencing is getting cheap.

Answer: Data privacy is crucial

Genomic data storage is expensive

Question: Any way to fix that?

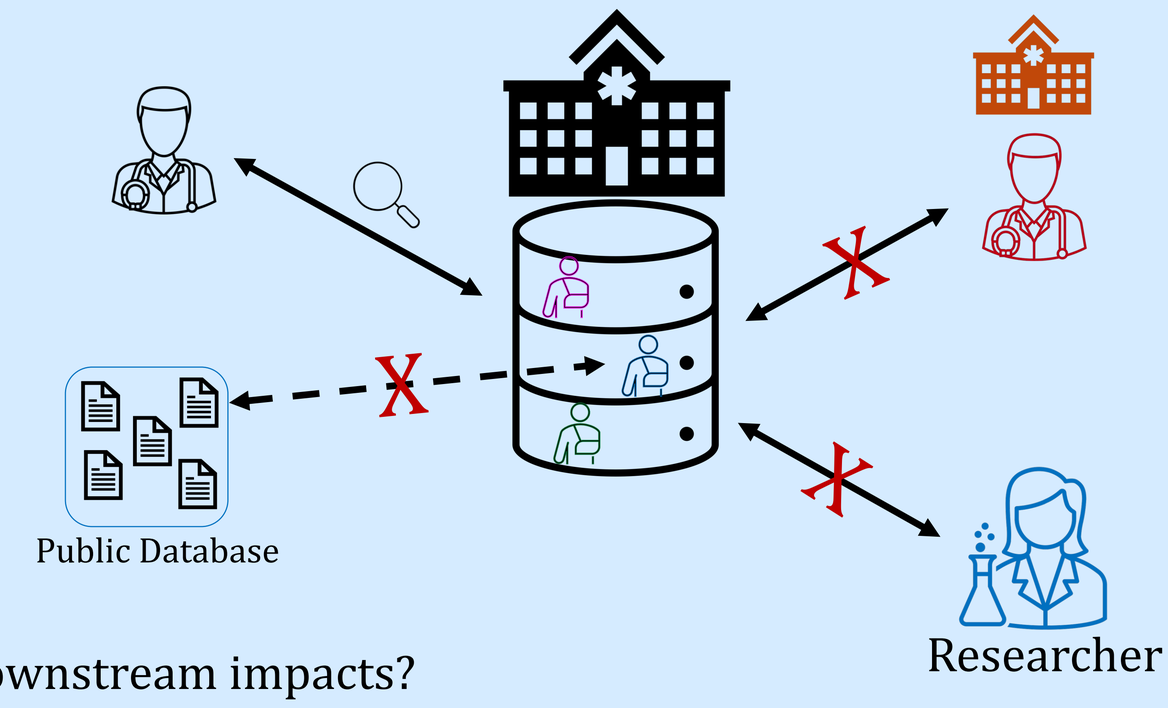
Personal Genome Sequence Data

VCF Variant Call Format (VCF)

Very large (~GBs in size)

GDPR Require strict privacy protections

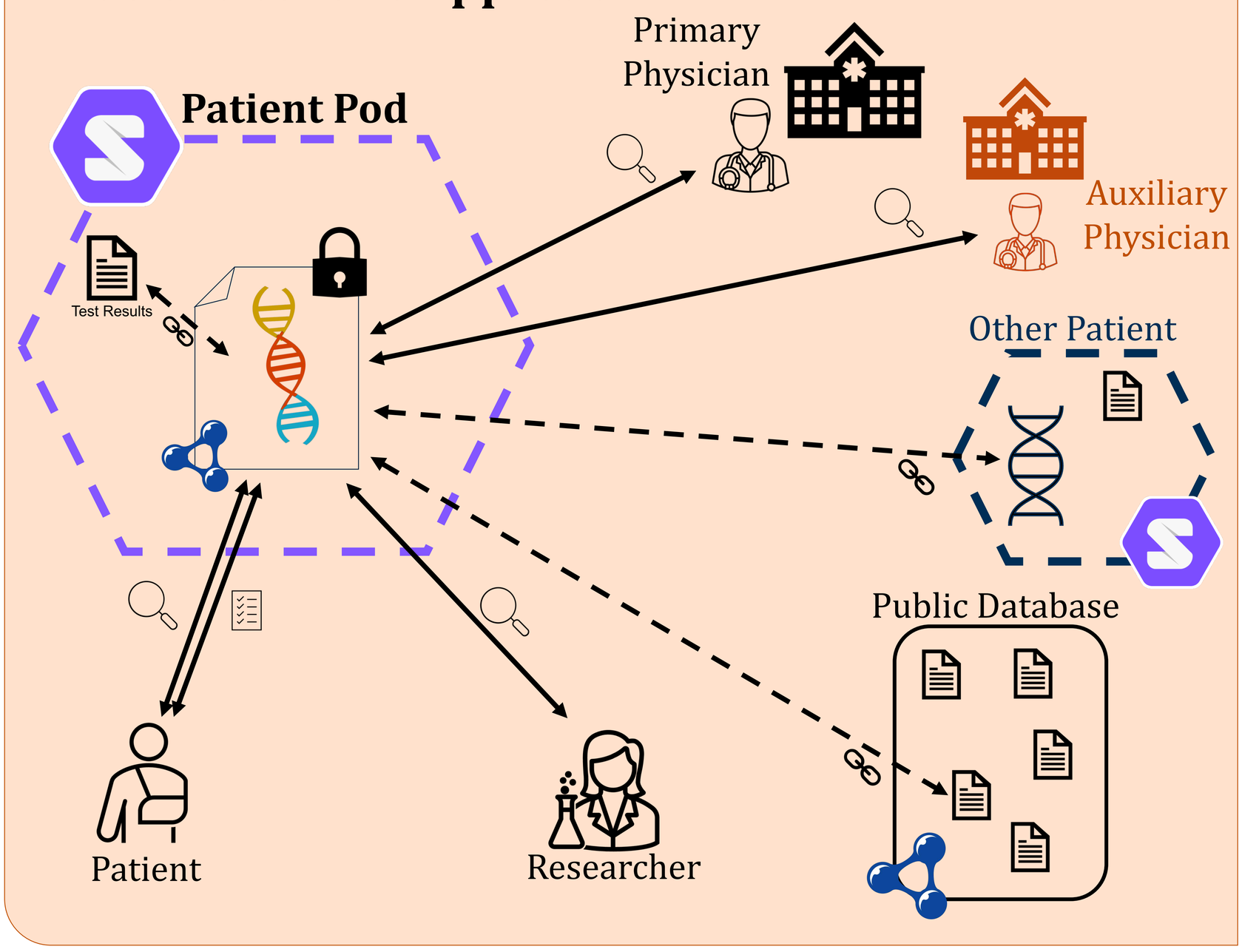
The current institution-centric system



Downstream impacts?

- Reduced data sharing
- Reduced patient transparency
- Increased pertinent data dispersion
- Data duplication

A Patient-centric Approach



Key Challenges

Privacy granularity

- Governance?
- How granular is adequate?

RDF Conversion + Data Linkages

- VCF to RDF Conversion?
- Automation of data linking?

Querying performance

- Many, large data sources
- Algorithms that use indexes?

Engineering Tasks

- Genomic data storage in Solid pods
- Data privacy management
- Data Sharing via the web

Research Pursuits

- Represent genomic data as RDF?
- Linking genomic and health data?
- Single and Many pod Querying?

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